STATS401 · GROUP 17 · DATA VISUALIZATION

What counts as an overtake?

Separating on-track passes from position changes in Formula 1, 2023–2025

43.3%
of recorded swaps are tagged as true on-track passes

01 · INTRODUCTION

One number, several kinds of event

Formula 1 often uses overtake totals to describe a race, but a recorded position change may result from a pit stop, retirement, neutralized period or post-race adjustment. We examine 2023–2025 data to separate live on-track passes from these other changes and explore their patterns through five linked visualizations.

19,303
recorded swaps

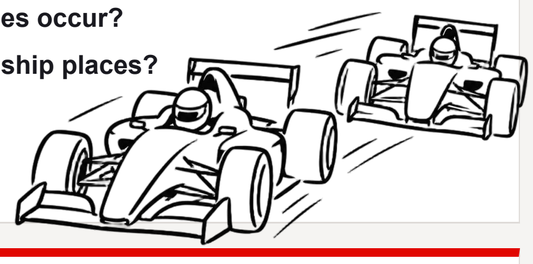
43.3%
on-track passes

46.2%
pit-related swaps

RESEARCH QUESTIONS

What does a pass reveal?

1. **How many recorded overtakes are genuine on-track passes?**
2. **When, where and between whom do passes occur?**
3. **Do frequent passers gain more championship places?**



02 · DATA & METHODS

From raw records to five linked views

OpenF1 provides overtakes, drivers, laps, pit stops, race control and car locations; Jolpica provides after-round standings. We analyse 70 races and 18 sprints from 2023–2025.

1 · Collect
OpenF1 + Jolpica
88 race / sprint sessions

2 · Clean & join
Align event, lap, driver and track records
19,303 valid swaps

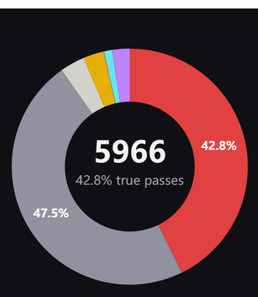
3 · Classify
One of seven exclusive event categories per swap

4 · Explore
Five interactive D3.js views with shared filters

03 · CORE RESULT · COMPOSITION

What share is a true pass?

The ring divides each session or season into seven exclusive event types. Its centre reports the total and on-track share; selecting a slice on the website filters the related views so users can compare definitions.



TAG	SWAPS	SHARE
True pass	2551	42.8%
Pit	2831	47.5%
Retired	202	3.4%
Neutralized	169	2.8%
Unlapping	4	0.1%
Formation	65	1.1%
Post-race	144	2.4%

Key insight. Across 2023–2025, only 43.3% of 19,303 swaps were on-track passes, while 46.2% were pit-related. The pass share stayed close to 43% each year: 44.0%, 42.8% and 43.0%, respectively.

INTERACTIVE VISUALIZATION SYSTEM

One dataset, five perspectives

Our D3.js website connects the composition ring with a driver network, lap strip, ranking chart and circuit map.

Choose context · Select a season and race or sprint.

Focus the field · Filter a driver or team across all five views.

Change the definition · Select an event category to update related charts.

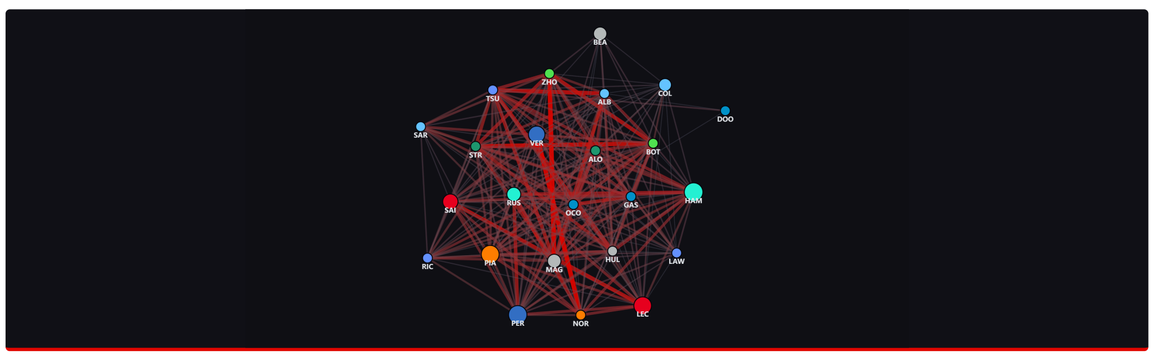
Inspect details · Use tooltips, the lap slider and map zoom.

Scan to explore the live dashboard
Five original interactive views

VIEW 1 · PAIR STRUCTURE

Who passes whom?

The force-directed network turns each driver into a team-coloured node, sized by net overtakes. A link joins every pair that exchanged positions; its thickness reflects the exchange count among the pairs currently shown. Filters let us test whether a rivalry persists under a stricter pass definition.

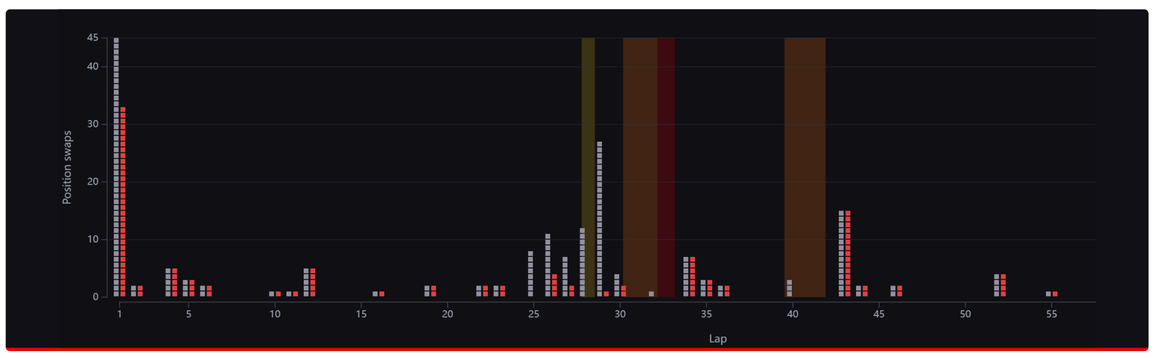


Reading the figure. With true passes selected, 2,551 events connect 24 drivers across 230 pairs at the two-exchange threshold in 2024. Verstappen–Norris leads with 33 exchanges (1.3%), yet the many other links show that passing extends across the field.

VIEW 2 · TIMING

When do the swaps happen?

This event strip places each recorded swap as a square on the race-lap axis. Separate raw and true-pass rows reveal how the pattern changes after tagging, while shaded bands mark safety-car, virtual safety-car and red-flag periods that can create misleading position changes.

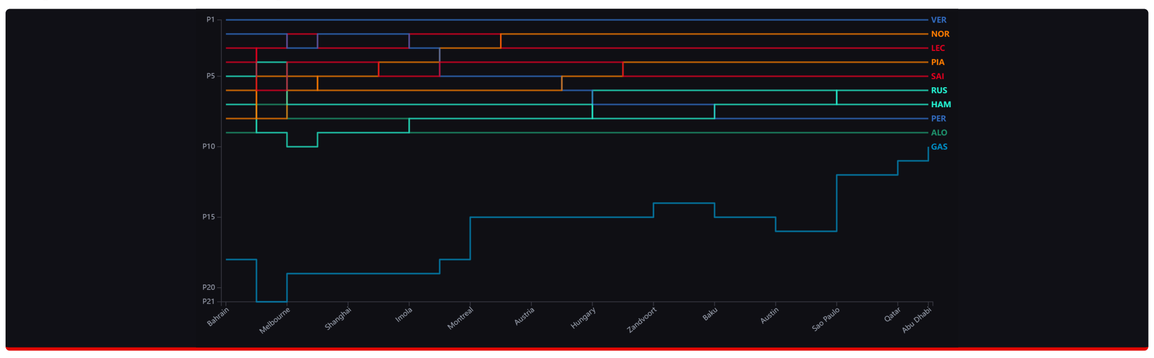


Reading the figure. At São Paulo 2024, 182 of 183 swaps can be assigned a lap and 105 qualify as true passes. The raw row peaks at the start and within neutralized bands; the cleaned row removes much of that activity but retains genuine Lap 1 passing.

VIEW 3 · RANK

Do frequent passers gain places?

Step lines follow a driver's championship position after each round; this print view displays the top ten 2024 finishers. We compare these season-long rank paths with each driver's net passes to ask whether overtaking frequently translates into a higher place in the standings.

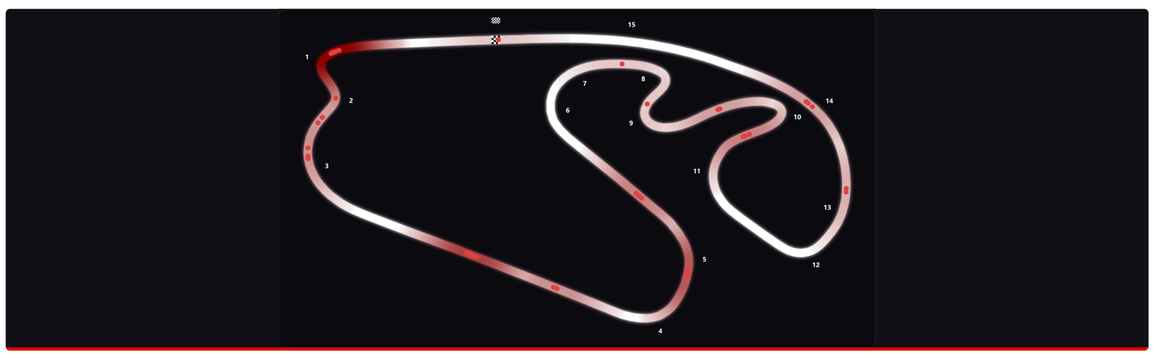


Reading the figure. Hamilton led in net passes (+51) but finished seventh, unchanged from Round 1; Pérez was +49 yet fell from second to eighth. Gasly gained eight championship places despite a negative net pass count (−10). Passing frequency does not predict rank change on its own.

VIEW 4 · LOCATION

Where on the circuit?

We reconstruct the circuit from one racing lap of car-location samples, then place each selected swap at the overtaking car's coordinates. Corner numbers supply spatial landmarks. The map asks whether an early-lap spike in the timing view is concentrated at Turn 1 or spread around the circuit.



Reading the figure. Dots mark the selected São Paulo position changes at their recorded locations. Category and lap filters on the live map let visitors test whether first-lap activity is concentrated at Turn 1 or distributed across several parts of the circuit.

CONCLUSION

Definitions change the story

Only about 43% of recorded swaps were genuine on-track passes. Passing was distributed across driver pairs, and frequent passing did not guarantee a championship rank gain. Five linked views make these differences visible.

DISCUSSION & LIMITATIONS

Interpret with care

Although our five visualizations share driver and team filters, some chart-specific controls remain independent, making cross-view comparisons less convenient. In addition, dense connections in the network graph and overlapping points on the circuit map can create visual clutter, making individual overtakes harder to identify.

EXPLORE & DISCUSS

Which swaps would you count?

Try the interactive dashboard and tell us which chart changes your interpretation. Sources: OpenF1 timing and location data; Jolpica standings.

jeromewang6066.github.io/stats401-FinalProject-Group17

Chengzhi Sun · Tianyao Wang · Jietong Zhou